FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II

PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	C	24.	C	47.	A
2.	A	25.	A	48.	D
3.	A	26.	A	49.	B
4.	BD	27.	ACD	50.	A,D
5.	ACD	28.	ABC	51.	BCD
6.	ACD	29.	ABC	52.	BD
7.	AC	30.	BCD	53.	ABC
8.	AC	31.	BD	54.	D
9.	D	32.	B	55.	A
10.	C	33.	C	56.	B
11.	3	34.	1	57.	5
12.	4	35.	6	58.	4
13.	2	36.	4	59.	2
14.	5	37.	5	60.	0
15.	4	38.	3	61.	4
16.	3	39.	3	62.	6
17.	1	40.	6	63.	7
18.	9	41.	4	64.	6
19.	1	42.	6	65.	7
20.	9	43.	6	66.	7
21.	00031.64	44.	00252.00	67.	00375.00
22.	00024.75	45.	00167.20	68.	00000.32
23.	00048.28	46.	00009.175	69.	00031.00

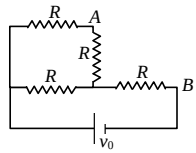
Physics

PART – I

SECTION – A

1. C

2. A
In steady state current will not flow from the capacitor.
If emf of battery is v_0, then potential difference between *A* and *B*.



$$V_A - V_B = \frac{0.4v_0}{5} = 12 \text{ V} \quad \Rightarrow \quad v_0 = 15 \text{ V}$$

3. A

$$V = iR \Rightarrow R = \frac{V}{i}, \Delta R = \left[\frac{\Delta V}{V} + \frac{\Delta i}{i}\right] \therefore \quad R = (4 \pm 0.012)\Omega$$

4. BD
5. ACD
6. ACD
7. AC
8. AC
9. D

10. C
Using IAOR

$$\text{mg}\frac{\ell}{2}\sin 53° = mg\frac{\ell}{2}\sin 37° + \frac{1}{2}I\omega^2$$

$$\omega = \sqrt{\frac{3}{5}} \Rightarrow V_{cm} = 5\sqrt{\frac{3}{5}} = \sqrt{15}$$

SECTION – C

11. 3

$$\frac{u^2}{g[1-\sin\beta]} = \frac{3u^2}{g[1+\sin\beta]} \Rightarrow 1+\sin\beta = 3-3\sin\beta \Rightarrow \sin\beta = \frac{1}{2} \Rightarrow \beta = 30^0$$

12. 4

$$220V = \sqrt{(176)^2 + V_0^2} \Rightarrow V_0 = \sqrt{(220)^2 - (176)^2} \; V = 132 \text{ V}$$

13. 2
14. 5
15. 4

$$\text{As } \omega = \frac{V_{rel}}{r} = \frac{5-3}{1} = 2\text{rad/sec}$$

$$T + 2\left(\frac{T}{2}\right) = \frac{2\times 2^2}{1} \Rightarrow T = 4\text{Newton}$$

16. 3
17. 1
18. 9
19. 1
20. 9

SECTION – D

21. 00031.64

22. 00024.75

$$f = \frac{3v}{4\ell} = 1000 \Rightarrow \ell = 24.75\,cm$$

23. 00048.28

Chemistry

PART – II

SECTION – A

24. C
(I) is trans and (II) is cis form

25. A

$$\underbrace{2KI}_{\text{Four ions}} + HgI_2 \longrightarrow \underbrace{K_2[HgI_4]}_{\text{three ions}}$$

26. A
Due to back bonding (A) is a weaker base.

27. ACD

28. ABC

29. ABC

30. BCD

31. BD

32. B

$$P_4(s) \longrightarrow P_4(g) \qquad \Delta H = 59KJ$$

$$P_4(s) \longrightarrow 4P(g) \qquad \Delta H = 1265\,KJ$$

$$P_4(g) \longrightarrow 4P(g) \qquad \Delta H = 1206\,KJ$$

Average $P-P$ bond enthalpy $= \dfrac{1206}{6} = 201KJ$

33. C
On an average, one $P-P$ bond of tetrahedral is broken and one new $P-P$ bond joining the two tetrahedral units is formed.

$201 - x = -104$

$\therefore x = 305\,KJ\,mol^{-1}$

SECTION – C

34. 1
It is orbital

35. 6

Isotopic no $= n - p = (58-26) - (26) = 32 - 26 = 6$

36. 4

$$X_{N_2} = \frac{P'_{N_2}}{K_H} = \frac{5\times 0.8}{1\times 10^5} \qquad (P'_{N_2} = P_{air} \times \text{mole fraction of } N_2)$$

$$= 4\times 10^{-4}$$

$$\therefore \quad \frac{n_{N_2}}{n_{N_2} + n_{H_2O}} = 4\times 10^{-5} = \frac{n_{N_2}}{n_{N_2} + 10} \qquad \left(n_{N_2} <<< 10\right)$$

$$\therefore \quad n = 4\times 10^{-4}$$

37. 5

$$P_S = \left(\frac{80}{100}\right)P^0, \quad w = ?, \quad m = 40$$

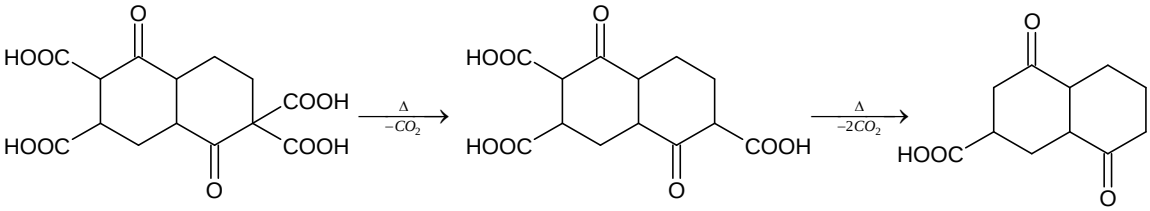
$$W = 114g, \quad M_{C_8H_{18}} = 114$$

$$\frac{P^0 - P_s}{P_s} = \frac{w \times M}{m \times W}$$

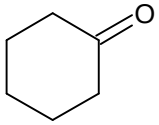
Or $$\frac{P^\circ - \left(\frac{80}{100}\right)P^\circ}{\left(\frac{80}{100}\right)P^\circ} = \frac{w \times 114}{40 \times 114}$$

$$w = 10g$$

38. 3

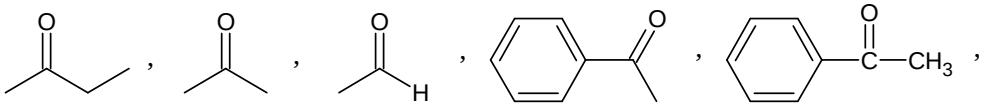


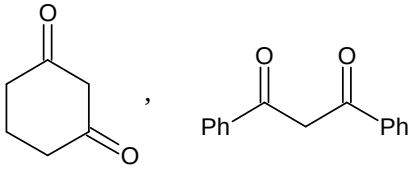
39. 3

Product: 

40. 6

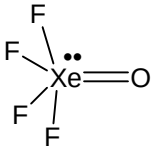
These compounds will give iodoform test





41. 4

$XeOF_4$



42. 6

The compound is $HClO_2$

43. 6

Azimuthal quantum number is zero for s-subshell

SECTION – D

44. 00252.00

$$\frac{r_1}{r_2}=\sqrt{\frac{M_2}{M_1}}\times\frac{P_1}{P_2}$$

Or $\frac{n_1}{t_1}\times\frac{t_2}{n_2}=\sqrt{\frac{M_2}{M_1}}\times\frac{P_1}{P_2}$

Or $\frac{1}{38}\times\frac{57}{1}=\sqrt{\frac{M}{28}}\times\frac{0.8}{1.6}$

$\therefore \quad M=252$

45. 00167.20

Given,

$$\frac{1}{2}H_2(g)+(Aq)\rightarrow H^+(aq);\qquad \Delta H^\circ=0.....(\mathrm{i})$$

$$\frac{1}{2}H_2(g)+\frac{1}{2}Cl_2(g)\rightarrow HCl(g);\qquad \Delta H^\circ=-92.4kJ.....(\mathrm{ii})$$

$$HCl(g)+H_2O(l)\rightarrow H_3O^+(aq)+Cl^-(aq);\qquad \Delta H=-74.8kJ.....(\mathrm{iii})$$

By inspection method: $\mathrm{Eqs.}(\mathrm{ii})+(\mathrm{iii})-(\mathrm{i})$ reveals that

$$\frac{1}{2}Cl_2(g)+(Aq)\rightarrow Cl^-(aq);\qquad \Delta H=-167.2kJ$$

$i.e.$ Heat of formation of $Cl^-(aq)=-167.2kJ$

46. 0000 9.175

Let solubility of $PbBr_2$ is $S\ mol\ litre^{-1}$

$$PbBr_2(s)+Aq\rightleftharpoons PbBr_2(aq)\rightleftharpoons \underset{\frac{S\times80}{100}}{Pb^2}+\underset{\frac{2S\times80}{100}}{2Br^-}$$

Since, $PbBr_2$ ionises to 80% only.

Now $\quad K_{sp}=\left[Pb^{2+}\right]\left[Br^-\right]^2$

Or $\quad 3.2\times10^{-5}=\left[\frac{S\times80}{100}\right]\left[\frac{2S\times80}{100}\right]^2$

$$S=0.025\,mol\,litre^{-1}$$

$$=0.025\times367\,g\,litre^{-1}$$

$$=19.175\,g\,litre^{-1}$$

Mathematics

PART – III

SECTION – A

47. A

$\left(\sin^{-1}\sin x - 1/2\right)^2 - 1/4$

48. D

$\left(x-2y-4\right)^2 = 12\left(2x+y\right)$

49. B

50. A,D

knowing the volume of the prism we find its altitude H = (A A_1) = and designating the vertex A_1 (x_1, y_1, z_1) relate the co-ordinates of the vector $\vec{AA}_1$ = (x - 1, y, z - 1) and its length. We get the other equation from the condition $\vec{AA}_1$ perpendicular to $\vec{AC}$]

51. BCD

52. BD

53. ABC

54. D

55. A

For $\lambda = 10 \Rightarrow t_1 = 2,\ t_2 > 2$ and if $x + 1/x = 2 \Rightarrow x = 1$ and $x + 1/x = t_2$ so there will be two distinct values of x

56. B

Since domain of $f\left(t\right) = 2t^3 - 9t^2 + 30,\ t = x + 1/x,\ |t| \geq 2\ f\left(-2\right) = -22,\ f\left(2\right) = 10,$ critical points at $t = 0\ \&\ 3;\ f\left(3\right) = 3$

SECTION – C

57. 5

58. 4

59. 2

60. 0

61. 4

62. 6

$$\lim_{x\to 0}\frac{\ln\cos 2x}{x^2} = -\frac{2^2}{2}, \lim_{x\to 0}\frac{\ln\cos 4x}{x^2} = -\frac{4^2}{2}$$

63. 7

64. 6

65. 7

66. 7

SECTION – D

67. 375.00

68. 00000.32

69. 00031.00